Investigation: PETROLEO BRASILEIRO S.A. PETROBRAS (CVM code 9512), fiscal year 2012, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Petróleo e Gás (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 1, filed on 2023-03-30): PETR3 (Ações Ordinárias, Bolsa, since 2018-05-14); PETR4 (Ações Preferenciais, Bolsa, since 2018-05-14).

Document used

2012 DFP, version 1, filed on 2013-02-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2013-02-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=24066

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.591 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 45 (n = 364)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.892 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 29 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.255 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 84 (n = 471)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.919 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 22 (n = 468)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.152 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value

Percentile in group and year: 60 (n = 472)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.995 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 57 (n = 383)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.959 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 37 (n = 465)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.049 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 47 (n = 487)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 1.084 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 69 (n = 480)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −5.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 32.5 days)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 28 (n = 440)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −2.21 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 11 (n = 439)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.052 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 48 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 2.522 (index)
Components: X1 = 0.072; X2 = 0.199; X3 = 0.046; X4 = 1.040 (indices)
Percentile in sector and year: 60 (n = 609); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (9 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): 1
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Highlighted in the following year's comparative: 2 (threshold of R$6,777.2 million, 1% of total assets of the version used). Values in R$ million:
BPP 2.03 Patrimônio Líquido Consolidado: 345,433.7 in the version used (v1, filed on 2013-02-04); 330,775.4 in the next DFP's comparative (v1, filed on 2014-02-25); difference of −14,658.3. Different value in the following year's comparative (restatement or reclassification).
BPA 1 Ativo Total: 677,716.3 in the version used (v1, filed on 2013-02-04); 669,031.6 in the next DFP's comparative (v1, filed on 2014-02-25); difference of −8,684.7. Different value in the following year's comparative (restatement or reclassification).
Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets

of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.
Full list: 42 accounts above the R$1,000 tolerance (group a, totals and key lines: 3; group b, reclassifications: 39), in the annex at the end.

What was known on the date

- DFP 2010, v1: 2011-02-25
- DFP 2010, v2: 2011-03-04
- DFP 2010, v3: 2011-03-04
- FCA 2012, v1: 2012-01-02
- FCA 2012, v2: 2012-01-27
- DFP 2011, v1: 2012-02-09
- DFP 2011, v2: 2012-02-14
- FCA 2012, v3: 2012-05-10
- DFP 2012, v1: 2013-02-04
- DFP 2013, v1: 2014-02-25

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da Petróleo Brasileiro S.A. - Petrobras em 31 de dezembro de 2012, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo nessa data, de acordo com as práticas contábeis adotadas no Brasil."
Auditor in the 2021 FCA: KPMG Auditores Independentes.
Auditor in the 2022 FCA: KPMG Auditores Independentes.
Auditor in the 2023 FCA: KPMG Auditores Independentes, KPMG Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: KPMG Auditores Independentes -> PricewaterhouseCoopers Auditores Independentes. The outgoing firm audited 6 consecutive fiscal years (2006 to 2011), by the start and end dates in the FCA.
- 2017: PricewaterhouseCoopers Auditores Independentes -> KPMG Auditores Independentes. The outgoing firm audited 5 consecutive fiscal years (2012 to 2016), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful

accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:38:45+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.

Annex: full list of differences in the following year's comparative (R$ million)

(a) BPP 2.03 Patrimônio Líquido Consolidado: 345,433.7 in the version used, 330,775.4 in the comparative (difference of −14,658.3)
(a) BPA 1 Ativo Total: 677,716.3 in the version used, 669,031.6 in the comparative (difference of −8,684.7)
(a) BPP 2.02 Passivo Não Circulante: 262,662.1 in the version used, 268,635.6 in the comparative (difference of 5,973.5)
(b) BPP 2.02.03.01.01 Imposto de Renda e Contribuição Social Diferidos: 39,261.6 in the version used, 0.0 in the comparative (difference of −39,261.6)
(b) DFC (indirect method) 6.02.01 Investimentos em Exploração e Produção: −41,933.4 in the version used, −80,033.1 in the comparative (difference of −38,099.7)
(b) DFC (indirect method) 6.02.02 Investimentos em Abastecimento: −26,932.1 in the version used, −284.9 in the comparative (difference of 26,647.2)
(b) BPP 2.02.04 Provisões: 40,830.3 in the version used, 61,593.2 in the comparative (difference of 20,762.9)
(b) BPP 2.02.04.02 Outras Provisões: 38,245.1 in the version used, 59,008.1 in the comparative (difference of 20,762.9)
(b) BPP 2.02.04.02.04 Planos de Pensão e de Saúde: 18,952.9 in the version used, 39,715.9 in the comparative (difference of 20,762.9)
(b) BPP 2.02.03 Tributos Diferidos: 39,261.6 in the version used, 24,472.2 in the comparative (difference of −14,789.4)
(b) BPP 2.02.03.01 Imposto de Renda e Contribuição Social Diferidos: 39,261.6 in the version used, 24,472.2 in the comparative (difference of −14,789.4)
(b) BPP 2.03.06 Ajustes de Avaliação Patrimonial: 50.5 in the version used, −14,453.6 in the comparative (difference of −14,504.1)
(b) BPA 1.02 Ativo Não Circulante: 559,614.5 in the version used, 550,929.7 in the comparative (difference of −8,684.7)
(b) BPA 1.02.01 Ativo Realizável a Longo Prazo: 47,215.2 in the version used, 38,530.5 in the comparative (difference of −8,684.7)
(b) BPA 1.02.01.06 Tributos Diferidos: 21,965.8 in the version used, 13,281.1 in the comparative (difference of −8,684.7)

(b) BPA 1.02.01.06.01 Imposto de Renda e Contribuição Social Diferidos: 11,292.8 in the version used, 2,608.0 in the comparative (difference of −8,684.7)
(b) BPP 2 Passivo Total: 677,716.3 in the version used, 669,031.6 in the comparative (difference of −8,684.7)
(b) BPA 1.02.01.09.04 Investimentos em Empesas Privatizáveis: 0.0 in the version used, 5,509.5 in the comparative (difference of 5,509.5)
(b) BPA 1.02.01.09.06 Adiantamento a Fornecedores: 6,448.5 in the version used, 1,101.6 in the comparative (difference of −5,346.9)
(b) DFC (indirect method) 6.01.01.08 Despesa Atuarial com Plano de Pensão e Saúde: 4,074.4 in the version used, 16.3 in the comparative (difference of −4,058.1)
(b) DFC (indirect method) 6.01.01.07 Valor Resid. de Bens Perm. Baixados: 80.2 in the version used, 4,074.4 in the comparative (difference of 3,994.2)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: 67,010.3 in the version used, 64,912.1 in the comparative (difference of −2,098.2)
(b) DFC (indirect method) 6.01.01.10 Imposto Renda e Contrib. Soc. Dif. Líq.: 4,256.3 in the version used, 2,221.9 in the comparative (difference of −2,034.4)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: −8,297.0 in the version used, −6,262.6 in the comparative (difference of 2,034.4)
(b) DFC (indirect method) 6.01.02.04 Impostos, Taxas e Contribuições: −2,340.6 in the version used, −306.3 in the comparative (difference of 2,034.4)
(b) DRE 3.08.01 Corrente: −2,537.9 in the version used, −4,572.2 in the comparative (difference of −2,034.4)
(b) DRE 3.08.02 Diferido: −4,256.3 in the version used, −2,221.9 in the comparative (difference of 2,034.4)
(b) BPP 2.02.02.02.03 Outras Contas e Despesas a Pagar: 1,395.3 in the version used, 0.0 in the comparative (difference of −1,395.3)
(b) BPP 2.01.03.01.01 Imposto de Renda e Contribuição Social a Pagar: 1,862.4 in the version used, 704.3 in the comparative (difference of −1,158.1)
(b) BPP 2.01.03.01.02 Outras Obrigações Fiscais Federais: 7,471.3 in the version used, 8,629.4 in the comparative (difference of 1,158.1)
(b) BPA 1.02.01.09.05 Depósitos Judiciais: 5,509.5 in the version used, 6,448.5 in the comparative (difference of 939.0)
(b) DFC (indirect method) 6.02.08 Recebimentos pela venda de ativos (desinvestimentos): 0.0 in the version used, 569.0 in the comparative (difference of 569.0)
(b) BPA 1.01.08.01 Ativos Não-Correntes a Venda: 0.0 in the version used, 289.9 in the comparative (difference of 289.9)
(b) BPA 1.01.08.03 Outros: 3,662.6 in the version used, 3,372.7 in the comparative (difference of −289.9)
(b) BPA 1.01.08.03.03 Outros: 1,768.0 in the version used, 1,478.1 in the comparative (difference of −289.9)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: 0.0 in the version used, −154.1 in the comparative (difference of −154.1)
(b) DFC (indirect method) 6.01.02.03 Fornecedores: 1,974.6 in the version used, 2,115.1 in the comparative (difference of 140.5)
(b) DFC (indirect method) 6.01.02.06 Oper. Curto Prazo com Subsid/Contr/Colig: 90.2 in the version used, 0.0 in the comparative (difference of −90.2)
(b) DFC (indirect method) 6.01.03.02 Outros Passivos: −517.1 in the version used, −453.3 in the comparative (difference of 63.8)
(b) DFC (indirect method) 6.01.03 Outros: −4,567.6 in the version used, −4,503.8 in the comparative (difference of 63.8)
(b) DFC (indirect method) 6.01.02.01 Contas a Receber: −3,017.5 in the version used, −3,067.7 in the comparative (difference of −50.2)
(b) DFC (indirect method) 6.01.03.01 Outros Ativos: −4,050.5 in the version used, −4,050.5 in the comparative (difference of −0.0)